

Religious Identity and Sexual-Orientation Discrimination: Experimental Evidence from Russia*

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Abstract

This paper examines how the salience of religious identity causally affects discrimination against sexual minorities. I run an online experiment with 2,093 Russians, in which I use a priming technique to activate participants' religious identities and measure discrimination through incentivized allocation tasks and LGB-related policy questions. The descriptive analysis shows that more religious respondents allocate fewer resources to same-sex profiles and show less support for LGB-related policies. Men, older respondents, and conservatives also discriminate more than other groups. In the causal analysis, I find that the religious prime has no statistically robust effect on either allocations or policy preferences, even among religious respondents. The results therefore add causal evidence to a literature that has largely relied on correlations, and show that although religiosity is related to anti-LGB views, a brief religious reminder has little effect on behavior or policy preferences.

JEL Classification: J15, J71, D91, Z12

Keywords: religious identity, sexual orientation, discrimination, experiment, priming

1 Introduction

The relationship between religious identity and discrimination against sexual minority groups is complex. Previous research provides mixed evidence, with some studies showing

*This study was supported by Charles University, GAUK project No.224121. I also acknowledge support from the Ministry of Education, Youth, and Sports (ERC CZ program, LL2303). The experiment in this study was pre-registered (AEA RCT Registry trial 8856). IRB approval was obtained from the CERGE Ethical Committee (UKCER/439220/2023). The experimental instructions are available in the Online Appendix.

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that higher religiosity is associated with negative attitudes towards sexual minorities (Burdette et al., 2005; Olson et al., 2006; Haider-Markel and Joslyn, 2008; Adamczyk and Pitt, 2009; Cunningham and Melton, 2013; Doebler, 2015; Gulevich et al., 2016; Suhay and Garretson, 2018), while others document a positive association between religiosity and acceptance of sexual minorities (Gouveia et al., 2020). Most studies in this area have an inherent correlational nature, because establishing a causal link between religiosity and discrimination based on sexual orientation is challenging. A primary methodological challenge is the difficulty of identifying exogenous variation in religious identities, as individuals self-select their religious affiliations, influenced by various observable characteristics, such as cultural background, family traditions, and personal experiences. This self-selection raises concerns about potential biases due to unobservable factors (Altonji et al., 2005; Gervais and Norenzayan, 2012b; Benjamin et al., 2016; Hoffmann et al., 2020).

To address these concerns, some studies exploit priming methods and document a causal link between religious identity and economic behavior, including prosocial behavior, cooperation, support for suicide attacks, hostility, and etc. (Atran and Norenzayan, 2004; Shariff and Norenzayan, 2007; Ginges et al., 2009; McKay et al., 2011; Schumann et al., 2014; Benjamin et al., 2016).

In this paper, I use a priming technique to create exogenous variation in the salience of religious identities to individuals affiliated with it, thus measuring the causal effect of the salience of religious identity on sexual-orientation discrimination¹. I measure subjects' discriminatory behavior through a series of monetary allocation tasks. Participants are randomly assigned to either a Religious-salient or Control condition. After a few demographic questions at the beginning of the experiment, those in the Control condition proceed directly to the allocation task stage, while those in the Religious-salient condition first respond to four religiosity-related questions, which serve as a prime for the salience of religious identity. Subsequently, participants are asked to allocate 500 RUB² between two profiles: one representing an individual with a same-sex partner (further, *same-sex profile*) and the other representing an individual with a different-sex partner (further, *different-sex profile*). Participants are also asked about their support for LGB-related policies³ - support for same-sex marriage and child adoption by same-sex couples.

The research question of this paper is: How does religious identity affect discriminatory

¹In this study, the majority of religious participants identify as Orthodox Christian, reflecting the religious composition of the Russian population. The sample also includes participants identifying with Islam (2.47%), Buddhism (0.86%), Protestantism (0.67%), and other religions (3.66%), as well as those with no religious affiliation (23.76%) and those who found it hard to answer (2.85%). While the analysis focuses on religiosity and religious affiliation in general, the findings are particularly relevant to contexts in which Orthodox Christianity is the predominant religion.

²RUB refers to the Russian ruble. At the time of the experiment, 500 RUB was approximately 5 euros.

³LGB stands for Lesbian, Gay, and Bisexual.

behavior and policy support towards individuals with same-sex partners? I hypothesize that activated religious identity will have a negative effect on sexual-orientation discrimination (pre-registered). This would mean that participants in the Religious-salient condition will discriminate against same-sex couples more than those in the Control condition. To answer the research question, I conducted an online experiment with a representative sample of Russian citizens by gender and age. Russia is a predominantly Orthodox country in which 66% of the population are members of the Russian Orthodox Church. The priming evokes the religious identity of participants, thereby facilitating an exploration of the causal relationship between religious identity salience - one particular channel through which religious identity may affect behavior - and discrimination based on sexual orientation⁴.

The results are as follows. A non-discriminatory allocation would be an equal split of 250 RUB between the two profiles. The data shows a clear departure from this benchmark: on average, participants allocate 77 RUB — or 27 percentage points of the total — less to same-sex profiles than to different-sex ones, and 13% choose not to allocate any funds to same-sex profiles at all. I find that religious affiliation is associated with reduced allocations to same-sex profiles and with lower support for related policies. Certain personal characteristics are also associated with stronger discriminatory behavior, in particular, being male, older, and politically conservative.

Next, I examine whether making religious identity salient affects discriminatory behavior and policy support. Treatment effects are negative for allocation outcomes and near zero for policy outcomes, but none are statistically significant — even when I restrict the sample to religious respondents. I also examine whether treatment effects vary with religiosity level. To do so, I use a continuous Religiosity Index constructed from respondents' reported religious affiliation, church attendance, prayer frequency, and the importance of religion in their lives; higher values indicate greater religiosity (see Section 2.5 for details). Based on respondents' religious affiliation and, among the affiliated, the value of this index, I divide respondents into three groups — Not Religious, Less Religious, and More Religious. The heterogeneity analysis shows that the negative effect of the prime on allocations to same-sex profiles and policy support is most pronounced among Less Religious respondents, while the Not Religious group shows no statistically significant response, and the More Religious group shows positive estimates for policy outcomes. However, differences across religiosity groups are not statistically significant for the allocation outcomes and, as discussed in Section 3.2, these subgroup patterns are exploratory and not all effects are robust to broader multiple-hypothesis correction

The causal relationship between religious identity and sexual-orientation discrimination has not been extensively explored. Several studies have examined the correlation between

⁴The method used in this study includes priming through direct questioning about religion and religious practices. Further details on this technique are provided in Section 2.

religiosity and attitudes toward homosexuality. Whitley (2009) offers a meta-analysis of sixty one studies that analyze the connection between religiosity and attitudes towards sexual minority groups. The author shows that most forms of religiosity are correlated in varying degrees to negative attitudes towards lesbians and gay men. Haider-Markel and Joslyn (2008) examine the relationship between perceptions regarding the origins of homosexuality and its primary determinants, with a special focus on religiosity and political ideology. They utilize the frequency of church attendance as their primary measure for religiosity and find that religiosity reduces the probability of associating homosexuality with biological origins. Over 50% of their respondents, and a significant 85% of regular church attendees, considered homosexual behavior to be sinful. Thus, while many religions advocate for tolerance towards different groups, several major world religions disapprove of homosexuality (Duck and Hunsberger, 1999; Toulouse, 2002; Lamontagne et al., 2018). Reasons for this might include religious belief systems, as well as non-doctrinal factors that may mediate the role of religiosity, e.g. beliefs about the nature of homosexuality, or homosexuality as a perceived threat, etc. (Duck and Hunsberger, 1999; Cottrell and Neuberg, 2005). Another study, Gouveia et al. (2020), investigates discrimination against same-sex couples in the Portuguese rental market and shows that an increase in parish-level religiosity is correlated with discrimination against male same-sex couples, though it exhibits more muted effects on discrimination against female same-sex couples. The authors suggest that in numerous religious communities, the concept of universal love predominates over negatively perceived moral attitudes.

This study is closely related to Baghumyan (2023), which also explores the underlying mechanisms of discrimination against same-sex couples in Russia. Both studies were conducted as part of the same experiment, which included two separate treatment arms and one shared control group. Importantly, the treatment arms do not interact with each other, and participants were randomly assigned to each treatment group. While Baghumyan (2023) focuses on the effects of biological attributions of homosexuality on discriminatory behavior, and suggests that exposure to evidence on the biological causes of homosexuality increases discriminatory behavior towards same-sex couples in Russia, the current study examines the impact of religious identity salience on discrimination.

Throughout this paper, I distinguish between three related but distinct concepts. *Religious identity* refers to self-categorization as a member of a religious group — the sense of belonging to a faith tradition as part of one’s social identity (Tajfel and Turner, 1981). *The salience of religious identity* refers to the temporary psychological prominence of that identity, induced experimentally by the prime (Benjamin et al., 2016; Cohn and Maréchal, 2016) — it is not a change in who a person is in a religious sense, but in how psychologically activated that aspect of one’s identity is at a given moment. *Religiosity* refers to the underlying level of religious affiliation, practice, and belief (Stark and Glock,

1968), as captured by respondents’ self-reports and summarized in the Religiosity Index. These three concepts play different roles in the analysis: the experiment identifies the causal effect of an increase in identity salience on discriminatory behavior, holding the underlying level of religiosity fixed. It does not address the effects of becoming permanently more religious, nor does it compare attitudes across religious denominations or doctrines. The heterogeneity analysis by religiosity groups is motivated by the idea that identity salience — the mechanism the prime operates on — may respond differently depending on a person’s level of religiosity (discussed in more detail in Section 3).

This paper contributes in several ways. First, it adds to the broader literature on discrimination against sexual minorities (e.g. Weichselbaumer, 2003; Carpenter, 2008; Drydakis, 2009, 2011; Tilcsik, 2011; Drydakis, 2014). Second, to the best of my knowledge, this is the first study that provides causal evidence of the effect of the salience of religious identity on sexual-orientation discrimination. Finally, I study sexual-orientation discrimination in a religious environment with strong and widespread anti-gay attitudes.

The paper proceeds as follows. In Section 2, I describe the experimental design, sample, and measures. In Section 3, I present the main results and discuss heterogeneity in treatment effects based on religiosity. Section 4 concludes.

2 Experimental Design

2.1 Sample

I conducted a pre-registered online experiment with a sample of $N = 2,093$ Russians via TGM Research, a market research platform. The sample matches the Russian population well in terms of gender, with 48% males, and age, with a median age of 42. It also represents different regions of the country and marital status fairly well. According to the latest wave of the World Value Survey, 69.7% of Russians identify as religious (Haerpfer et al., 2020), 49.6% report that religion plays a significant role in their lives, rating it as either very important or rather important. In this study’s sample, 74% of participants are religious, 66% identify as Russian Orthodox, and 24% report being non-religious. Appendix Table B.1 reports the main demographic characteristics of the sample and compares them to Census data, and Appendix Table B.2 presents the religion-related composition of the sample.

2.2 Attitudes towards sexual minorities in Russia

In 2013, Patriarch Kirill, the head of the Russian Orthodox Church, asserted that the legal recognition of same-sex marriage is "a very dangerous sign of the Apocalypse" (Herszenhorn, 2013). This declaration underscores the Church's role in reinforcing strong anti-gay sentiments within Russia. The Russian legal framework further exacerbates this climate because it lacks anti-discrimination laws aimed at protecting the labor rights of LGB individuals. In 2013, the Russian State Duma adopted a law to "protect children from information that promotes denial of traditional family values", referred to in English-language media as "the gay propaganda law" (Luhn, 2015). This absence of protective legislation, restrictive laws, and existing anti-gay attitudes may potentially intensify discriminatory practices against sexual minorities within Russia (Barron and Hebl, 2013; Pachankis and Bränström, 2018). According to the World Value Survey's 7th wave, 74% of Russians display low levels of acceptance toward homosexuality, 66% state they would not prefer homosexuals as neighbors, and 58% believe that homosexuality is never justifiable (Russian Public Opinion Research Center (VCIOM), 2021). Taken together, these sociocultural and legislative conditions make Russia a particularly interesting setting for studying discrimination against sexual minorities and the role of religious identity.

2.3 Measuring discrimination

I use a spectator design to measure discrimination against same-sex profiles, which is a common method in economics (Lane, 2016; Cappelen et al., 2016; Enke et al., 2021). This approach is similar to that in Baghumyan (2023). It involves five allocation tasks in which participants divide a fixed sum (500 RUB, or about 5 EUR) between two recipients using a slider. The main tasks require allocating funds between a male same-sex and a male different-sex profile, and, similarly, between a female same-sex and a female different-sex profile. In addition, participants completed three comparison tasks involving only different-sex profiles. Sexual orientation was signaled through the partner's name shown on each profile: a same-sex profile listed a partner name of the same gender as the recipient, whereas a different-sex profile listed a partner name of the opposite gender. Appendix Figure A.1 visualizes the allocation task between a male same-sex and a male different-sex profile⁵. The other tasks were constructed similarly, and all tasks were presented to participants in random order.

The profiles were based on information collected from real individuals recruited before

⁵All information presented in the profile is orthogonal to the type of partner — same-sex or different-sex — ensuring that any variation in the outcomes comes from the signal of sexual orientation rather than other information included in the profile.

the main experiment. To preserve anonymity while keeping the profiles realistic and incentive-compatible, I modified the names used in the profiles, including the partner’s name, while holding the remaining profile information fixed within each comparison. Participants were incentivized as follows: they were informed that, at the end of the experiment, 20 participants would be randomly selected, and one of their choices would be implemented. This design allowed the selected allocation decisions to be paid out to the real individuals behind the profiles.

To measure overall discrimination, I compare the amounts given to same-sex versus different-sex profiles. To assess the impact of religious identity salience on discrimination, I compare the average amounts allocated to same-sex profiles between Religious-salient and Control groups. The first set of outcomes includes the amount sent to male same-sex profiles, the amount sent to female same-sex profiles, and the average of these two amounts.

I also gathered data on participants’ views regarding LGB-related policies — recognizing same-sex marriages with the same rights as traditional marriages, and legalizing child adoption by same-sex couples. These views were elicited using a 5-point Likert scale ranging from “Disagree” to “Agree”. These two questions present the second set of outcome variables used in this study.

2.4 Treatment

The literature uses various priming techniques to examine the marginal impact of religiosity on economic and social behavior and preferences (Cohn and Maréchal, 2016). Some studies employ indirect priming methods, which may include tasks like sentence unscrambling with religious words or exposure to religious content through pictures and audio (Shariff and Norenzayan, 2007; Pichon et al., 2007; Gervais and Norenzayan, 2012a; Laurin et al., 2012a; Rounding et al., 2012). Other studies, including this study, utilize direct religious questioning (Ginges et al., 2009; Laurin et al., 2012b). In this approach, participants explicitly answer questions about their religious beliefs, practices, and the importance of religion in their lives. Shariff et al. (2016) provide a meta-analysis of studies that use religious priming for causal identification.

In this study, participants were randomly allocated into Religious-salient and Control conditions. The Religious-salient condition consists of four religiosity-related questions that are designed to activate participants’ religious identities. The Control condition does not include any priming. The religious module includes the following questions:

1. What is your religious preference?
2. How often do you attend church or a place of worship?

3. How often do you pray to God outside of religious services?
4. How important would you say religion is in your own life?

Because the goal of the treatment is to increase the salience of religious identity, the four religion-related questions are administered at different points in the experiment across conditions. In the Religious-salient condition, they appear before the allocation and policy tasks, serving as the prime; in the Control condition, the same questions are asked only at the end of the experiment, together with the remaining background questions. This timing difference is the experimental manipulation itself, and it is intentional — asking the religiosity questions after the tasks in the Control condition avoids inadvertently priming control respondents before they complete the main outcomes. This also means that the Religiosity Index is not measured at the same point in the survey across conditions — for treated respondents, it is recorded before the allocation tasks, while for control respondents, it is recorded after. As a result, the index may partly reflect the experience of the experiment itself rather than only stable underlying religiosity. This limitation is important to keep in mind when interpreting the heterogeneity analysis.

Appendix Tables B.3 and B.4 show that randomization was largely successful. The joint balance tests across variables indicate no systematic differences between groups. At the individual variable level, two variables show statistically significant differences in Table B.3 — being a student ($p = 0.082$) and having an acquaintance with homosexual orientation ($p = 0.067$). In Table B.4, a small number of individual category comparisons reach marginal significance — specifically, never attending church ($p = 0.031$), sometimes praying ($p = 0.044$), and the hard-to-answer and very important categories of religious importance ($p = 0.094$ and $p = 0.100$) — though the overall chi-square tests for each religion variable are not significant, and the joint balance test across all religion variables is far from significance. The absolute magnitudes of all individual differences are small and unlikely to alter the conclusions of the study, especially given the null average treatment effects reported in Section 3.2. The religion-related variables are well-balanced across treatment and control groups (Appendix Table B.4), suggesting that the Religiosity Index captures similar underlying distributions in both groups despite being measured at different points in the survey. In the main specification, I control for student status and having a homosexual acquaintance. The religion variables are not included as controls because, for the Religious-salient group they are measured before the allocation tasks and form the treatment itself — including them as covariates would risk controlling away the very mechanism the prime operates on.

2.5 Religiosity Index

The same four questions are also used to construct a continuous Religiosity Index for the heterogeneity analysis. The index is based on religious affiliation, church attendance, prayer frequency, and the self-reported importance of religion in respondents' lives. Following Anderson (2008), I combine these four components using inverse-covariance weighting, with higher values indicating greater religiosity. Appendix Table C.1 shows that these components are positively correlated with one another, and Appendix Figure C.1 presents the distribution of the index. In the heterogeneity analysis, I use this measure both as a continuous indicator of religiosity and to classify respondents into three groups: Not religious (raw affiliation = "No religion"), and among affiliated respondents, Less religious (index below the affiliated-sample median) vs More religious (index at or above the median).

3 Results

3.1 Descriptive Analysis

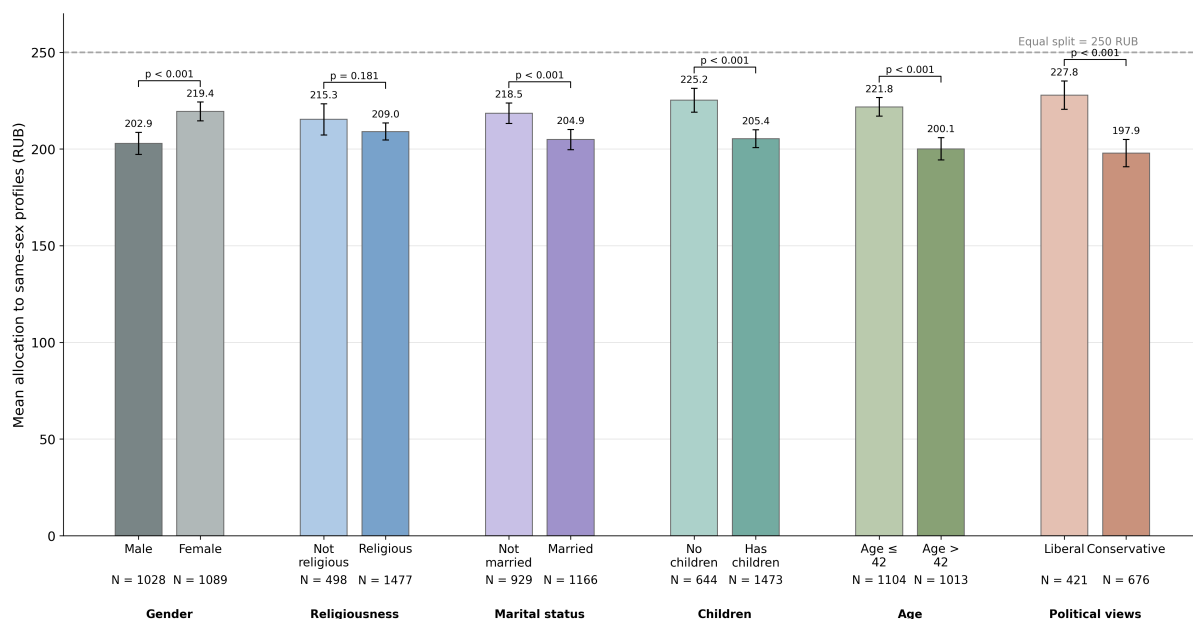
I start by documenting the prevalence of discrimination against same-sex profiles. The main task that I use to measure discrimination includes splitting 500 RUB between same-sex and different-sex profiles. Thus, discriminatory behavior against same-sex profiles would be one that includes an unequal split (i.e., less than 250 RUB) in favor of different-sex profiles. Based on this rationale, discriminatory patterns are evident in the data. On average, participants send 77 RUB, or 27 percentage points, less to same-sex profiles than they send to different-sex ones ($p < 0.001$).

Approximately 54% of participants allocate less than 250 RUB to same-sex profiles, and about 13% allocate exactly 0 RUB. Figure 1 shows clear subgroup heterogeneity in raw means: men, religious respondents, married respondents, parents, older respondents, and conservatives allocate less to same-sex profiles than other participants. Figure 2 reports subgroup differences after adding controls. Three patterns remain robust: women allocate more than men (diff = 9.25), $p = 0.026$; respondents above the median age allocate less than younger respondents (diff = -14.57), $p < 0.001$; and conservatives allocate less than liberals (diff = 22.70), $p < 0.001$. By contrast, adjusted differences by religious affiliation ($p = 0.514$), marital status ($p = 0.492$), and parenthood ($p = 0.954$) are small and not statistically significant⁶.

⁶Controls include gender, age, education and income level, marriage status, number of children, prior beliefs about the sources of discrimination (biological vs. other), sexual orientation, and having a homosexual acquaintance.

Appendix Figure D.1 takes a closer look at respondents who give 0 RUB to same-sex profiles. When controls are added, the picture is fairly consistent with what was observed above: men, older respondents, and conservatives are noticeably more likely to allocate exactly 0 RUB, while differences by religiosity and marital status largely disappear. The gender and political gaps are sharpest for male same-sex profiles — women are around 6.4 percentage points less likely than men to give nothing, and liberals are about 7.1 percentage points less likely to give 0 than conservatives. These gaps narrow somewhat for female same-sex profiles but do not disappear. Taken together, the pattern points to the same conclusion as the descriptive analysis: the most discriminatory responses tend to come from men, older participants, and those with conservative political views.

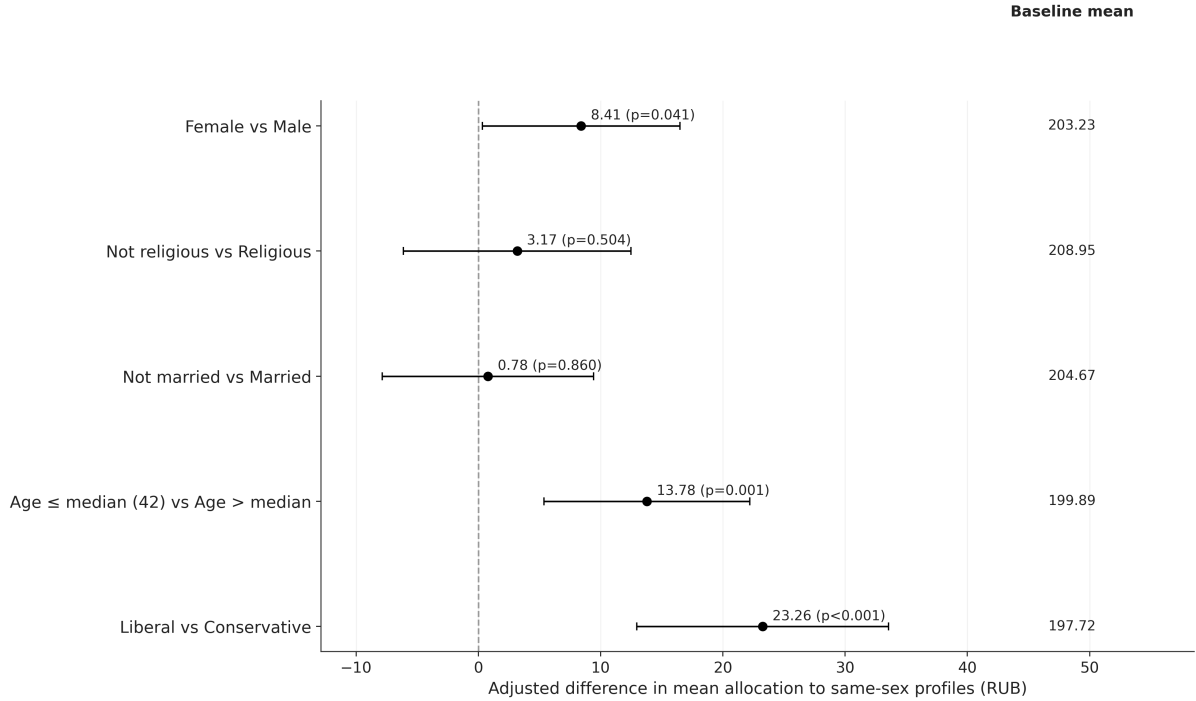
Figure 1: Mean allocations to same-sex profiles across respondent subgroups



Notes: The figure shows subgroup means of the average amount allocated to same-sex profiles, with 95% confidence intervals. The dashed horizontal line at 250 RUB marks the equal-split benchmark. P-values are based on raw pairwise comparisons of subgroup means within each subgroup.

There is also an interesting interaction between participants' gender and religiosity. Religiosity is presented by splitting the Religiosity Index into 3 groups: Not Religious, Less Religious, and More Religious. Figure 3 shows that, overall, male participants are more discriminatory than female participants. Across the groups of the Religiosity index, male allocations to same-sex profiles are very similar across groups (Not Religious: 201.6 RUB; Less Religious: 200.4 RUB; More Religious: 205.2 RUB), and differences among male religiosity groups are not significant (all FDR-adjusted $p > 0.68$). In contrast, the behavior of female participants differs by religiosity. In the Not Religious group, female participants allocate 232.2 RUB to same-sex profiles, which decreases by 23.3 RUB in the Less Religious group FDR-adjusted $p = 0.002$, and 10.5 RUB in the More religious group (FDR-adjusted $p = 0.212$).

Figure 2: Subgroup differences in allocations to same-sex profiles



Notes: Points are adjusted subgroup differences in mean allocation to same-sex profiles (RUB), estimated from OLS models with controls and HC3 robust standard errors; whiskers are 95% confidence intervals. The vertical dashed line marks zero effect. Labels report coefficient estimates and p -values. Baseline mean denotes the mean allocation for the reference subgroup in each comparison (Male, Religious, Married, Age > median, Conservative).

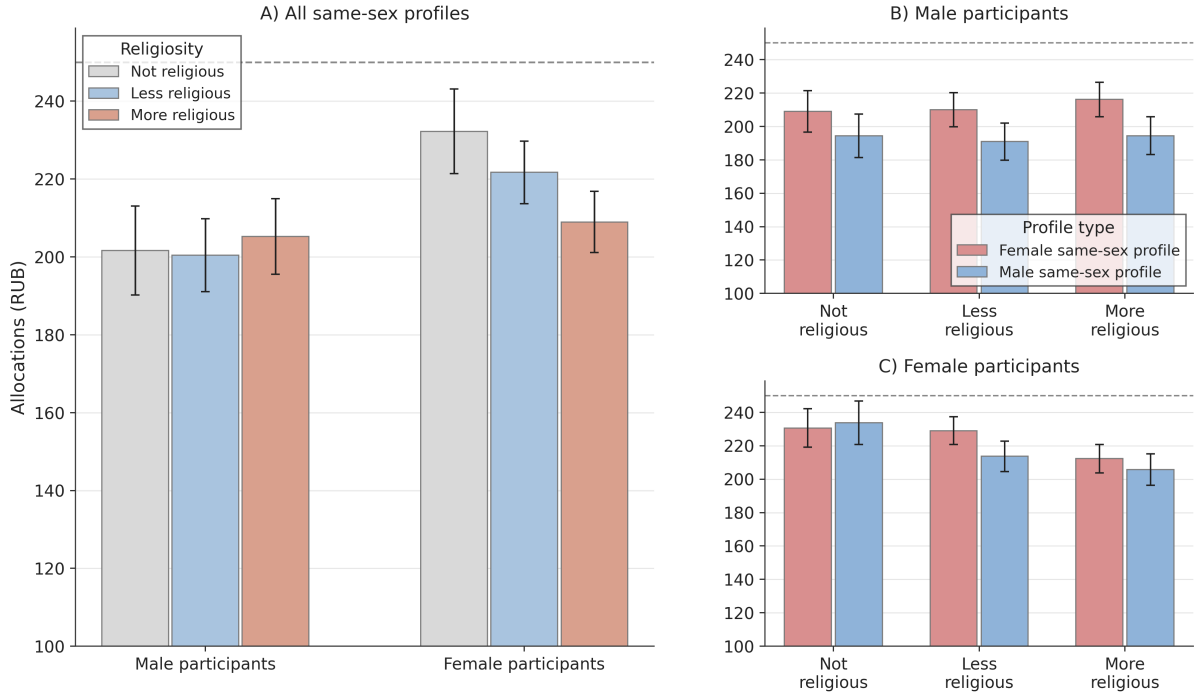
Figure 3 also reports allocations separately to male and female same-sex profiles. On average, male same-sex profiles receive lower allocations than female same-sex profiles. Among male respondents, allocations are quite stable across religiosity for both profile types. None of the religiosity-group differences among men is statistically significant.

Among female respondents, the pattern is different. Allocations decrease as religiosity increases. In the Not Religious group, female respondents allocate on average 233.7 RUB to male same-sex profiles, which drops to 205.7 RUB in the More Religious group (FDR-adjusted $p = 0.002$). When it comes to female same-sex profiles, female participants allocate an average of 230.6 RUB in the Not Religious group, which drops to 212.2 RUB in the More Religious group (FDR-adjusted $p = 0.031$).

In the Not Religious group, female respondents allocate an average of 233.7 RUB to male same-sex profiles. This amount decreases to 205.7 RUB in the More Religious group (FDR-adjusted $p = 0.002$).

In addition to allocation decisions, I also examine policy preferences: support for legal recognition of same-sex marriage and for child adoption by same-sex couples, measured on a 5-point Likert scale (1 = strong opposition, 5 = strong support). Figure 4 presents

Figure 3: Mean allocations to same-sex profiles by gender and religiosity



Notes: Bars show mean allocations (RUB) and whiskers show 95% confidence intervals. Religiosity groups are defined as follows: *Not religious* if the participant has no religious affiliation; among religious respondents, *Less religious* if the Religiosity Index is below the median, and *More religious* if it is at or above the median. The horizontal dashed line marks the equal-split benchmark at 250 RUB.

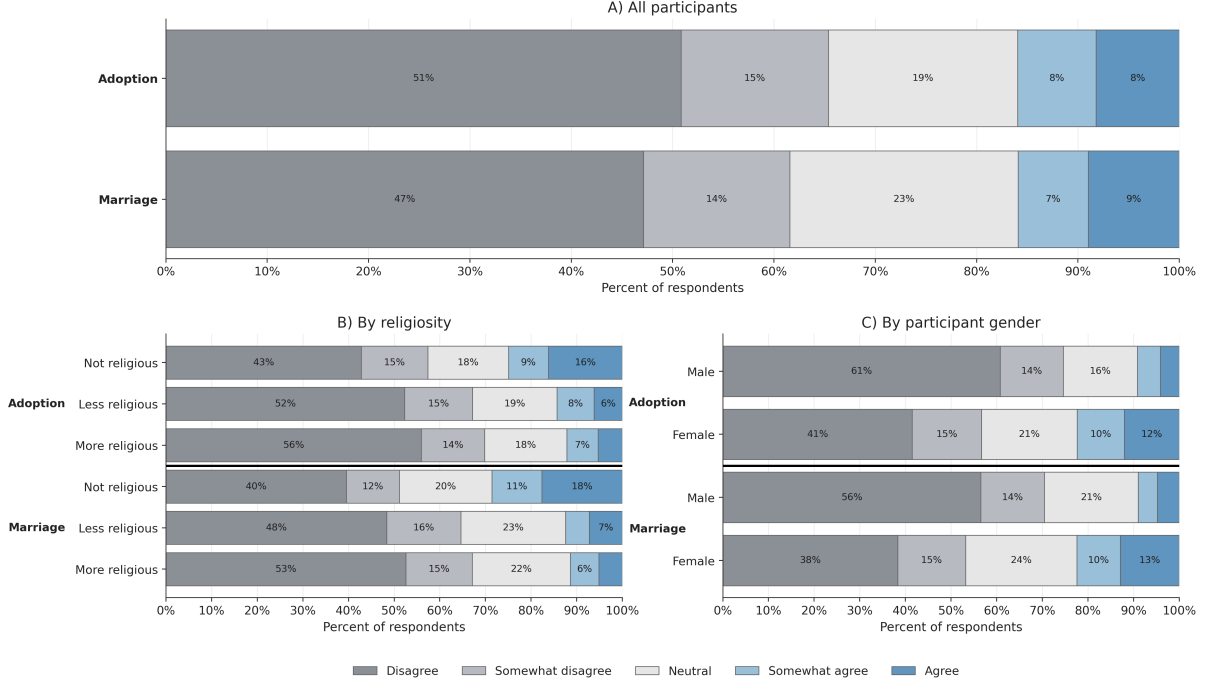
these distributions. Overall opposition is substantial: 66% of respondents either disagree or somewhat disagree with child adoption by same-sex couples, and 61% either disagree or somewhat disagree with legalizing same-sex marriage.

Figure 4 shows clear differences in policy support by religiosity and gender. In Panel B, opposition is higher among more religious respondents. For child adoption, the share who disagree or somewhat disagree is 58% for not religious respondents, 67% for less religious respondents, and 70% for more religious respondents. For same-sex marriage, the same pattern appears: 52%, 64%, and 68%.

Panel C shows a similar gap by gender. Men are more opposed than women to both policies. For child adoption, 75% of men vs 56% of women disagree or somewhat disagree. For same-sex marriage, the shares are 70% for men and 53% for women. Overall, support for pro-LGB policies is lower among more religious respondents and among men.

Table 1 reports Spearman correlations between the main outcomes and the Religiosity Index. The allocation outcomes are clearly related. All same-sex allocation is strongly correlated with male same-sex allocation ($\rho = 0.8472^*$) and female same-sex allocation ($\rho = 0.7032^*$). The correlation between male and female same-sex allocations is moderate ($\rho = 0.3876^*$).

Figure 4: Support for related policies by religiosity and participant gender



Notes: The bars show the distribution of participants for LGB-related policy support. Panel A shows the distribution across all participants. Panel B shows the distribution by religiosity groups. Panel C shows the distribution by participant gender.

Policy outcomes are also strongly related. Support for same-sex marriage and support for child adoption have a high correlation ($\rho = 0.8107^*$). Correlations between allocation outcomes and policy outcomes are positive but smaller ($\rho \approx 0.19$ to 0.25). This suggests they capture related, but not identical attitudes.

Higher religiosity is linked to lower support for same-sex policies ($\rho = -0.1532^*$ for marriage; $\rho = -0.1333^*$ for adoption). Higher religiosity is also linked to lower allocations to same-sex profiles ($\rho = -0.0600^*$, $\rho = -0.0419$, and $\rho = -0.0201$). Among allocation outcomes, only the pooled all same-sex measure is statistically significant. A likely reason is that the pooled measure is more precise than the separate male and female allocation measures.

3.2 Treatment Effects on the Main Outcomes

In this section, I examine the impact of the Religious-salient treatment on allocation decisions and policy support. Panel A of Table 2 shows that, in the full sample, treatment effects are negative for the allocation outcomes and close to zero for the policy outcomes, but none of the estimates is statistically significant. Because the priming treatment is designed to activate respondents' religious identities, any effect is more likely to be concentrated among religiously affiliated respondents. Panel B of Table 2 therefore reports

Table 1: Correlation between the main outcomes and the Religious Index

	Allocations to profiles			Support for policies		
	All same-sex	Male same-sex	Female same-sex	Same-sex marriage	Child adoption	Religiosity Index
All same-sex	1.0000					
Male same-sex	0.8472*	1.0000				
Female same-sex	0.7032*	0.3876*	1.0000			
Same-sex marriage	0.2509*	0.2463*	0.2041*	1.0000		
Child Adoption	0.2363*	0.2245*	0.1902*	0.8107*	1.0000	
Religiosity Index	-0.0600*	-0.0419	-0.0201	-0.1532*	-0.1333*	1.0000

Notes: The Religiosity Index follows Anderson (2008) and combines religious affiliation, church attendance, prayer frequency, and the importance of religion using inverse-covariance weighting. Correlations are Spearman rank correlations. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

treatment effects within the religious sample. The results consistently show negative effects across all allocation and policy outcomes. However, only the estimate for allocations to female same-sex profiles approaches statistical significance, and even this finding does not hold up after adjusting for multiple hypothesis testing. Taken together, the main outcomes provide little evidence of strong or reliable treatment effects when multiple comparisons are accounted for.

Table 2: Treatment effects on main outcomes

	Allocations to profiles			Support for policies	
	All same-sex (1)	Male same-sex (2)	Female same-sex (3)	Same-sex marriage (4)	Child adoption (5)
<i>Panel A: Full Sample</i>					
Religious-salient	-4.442 (3.770)	-3.319 (4.478)	-5.565 (4.045)	0.007 (0.052)	-0.018 (0.052)
N	2,074	2,074	2,074	2,074	2,074
Control mean	213.48	206.15	220.82	2.16	2.09
<i>Panel B: Religious Sample</i>					
Religious-salient	-6.180 (4.459)	-3.853 (5.346)	-8.507* (4.763)	-0.030 (0.059)	-0.055 (0.060)
N	1,463	1,463	1,463	1,463	1,463
Control mean	212.56	203.75	221.37	2.01	1.99

Notes: OLS estimates with HC3 robust standard errors in parentheses. Religious-salient is a treatment indicator equal to one for participants assigned to the religious-priming condition. Controls include gender, age, education, income, marital status, number of children, political views, prior beliefs about the biological causes of sexual orientation, sexual orientation, and having a homosexual friend. No religion-related controls are included. The religious sample in Panel B is defined using the paper's existing religious/non-religious classification. Higher values in columns 4–5 indicate greater support for same-sex marriage and child adoption by same-sex couples. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The null average treatment effects may partly reflect the fact that the prime is unlikely to operate uniformly across all respondents. The treatment is designed to activate religious identity, but this mechanism presupposes that respondents actually hold such an identity. For non-religious individuals, questions about religious beliefs and practices are unlikely to

trigger any meaningful change in identity salience — there is simply no religious identity to activate. At the other extreme, for highly religious individuals, religion may already be a very salient part of their self-concept, leaving little room for the prime to shift behavior further. This suggests that the prime is most likely to affect individuals for whom religion is meaningful but not constantly activated — that is, those who are moderately religious. Motivated by this reasoning, I examine whether the treatment effect varies systematically with respondents’ level of religiosity (not pre-registered).

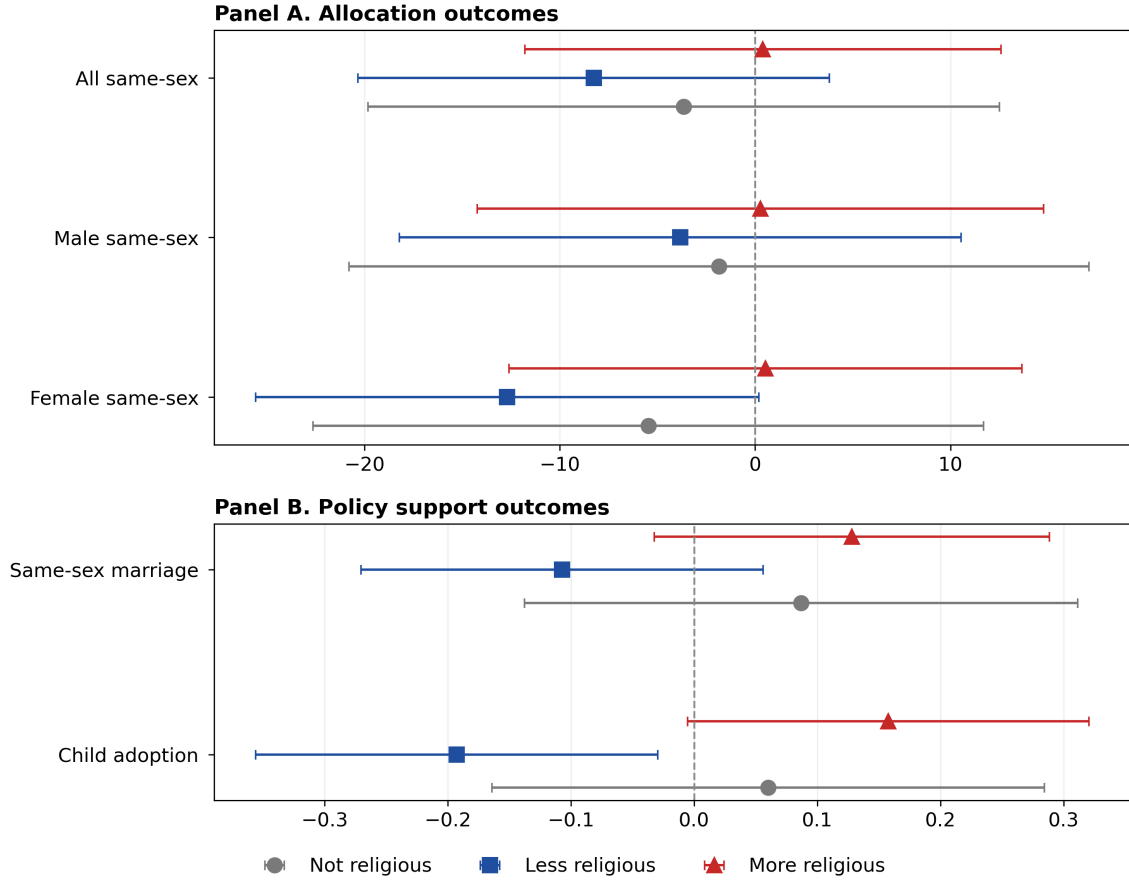
The treatment effect does not vary monotonically with religiosity. In the interaction models with treatment and the Religiosity index, the interaction term is not statistically distinguishable from zero, indicating no clear evidence that the treatment effect becomes steadily more negative as religiosity increases. I therefore turn to a more flexible grouped analysis. I divide respondents into three categories: Not Religious, Less Religious, and More Religious. The first group consists of respondents whose raw religious affiliation is “No religion.” Among religiously affiliated respondents, the latter two groups are defined by whether their religiosity index falls below or at or above the median of the affiliated sample.

Figure 5 shows the subgroup treatment effects across all five outcomes. In the Less Religious group, the prime decreases allocations for same-sex profiles and reduces support for related policies. In contrast, the More Religious group has estimated effects that are close to zero or even positive, and the Not Religious group does not show a clear response. The gap between the Less Religious and More Religious groups is largest when it comes to policy outcomes, as shown in Panel B.

Table 3 reports the grouped heterogeneity analysis and the corresponding formal interaction tests. As mentioned above, the most negative treatment effects appear among the Less Religious respondents. In this group, priming reduces allocations to female same-sex profiles by about 12.7 RUB (marginally significant) and lowers support for child adoption by same-sex couples by about 0.19 points on the five-point scale (significant at conventional levels). In contrast, the Not Religious group shows no clear treatment response across outcomes. Among the More Religious, allocation effects are close to zero, while policy coefficients are positive.

The formal tests in Table 3 show that statistically meaningful heterogeneity appears in policy outcomes rather than allocation outcomes. For allocations, interaction-based comparisons across religiosity groups are not statistically significant. For policy support, the differences are more pronounced: for same-sex marriage, the Less-versus-More Religious comparison is significant ($p = 0.044$), while for child adoption, both the Less-versus-More comparison ($p = 0.003$) and the joint interaction test ($p = 0.010$) are significant. The pattern is therefore consistent with the motivating idea that the prime may have a stronger

Figure 5: Treatment effects by religiosity subgroups



Notes: Dots are OLS estimates of the treatment effect of the Religious-salient vs Control, separately for each religiosity subgroup. Whiskers are 95% confidence intervals based on HC3 robust standard errors. Panel A reports allocation outcomes (in RUB); Panel B reports policy support outcomes (on a 1–5 Likert scale). The dashed vertical line marks zero effect. *Not religious* includes respondents whose raw religious affiliation is “No religion”; among affiliated respondents, *Less religious* and *More religious* are defined by whether the Religiosity Index falls below or at/above the median of the affiliated sample. Estimates are recovered as linear combinations from a pooled interaction model and correspond to Panel A of Table 3. Controls include gender, age, education, income, marital status, number of children, political identity, prior beliefs about the biological causes of sexual orientation, sexual orientation, and having a homosexual acquaintance.

influence among moderately religious respondents — it is the Less Religious, not the More Religious or the non-religious, who show a stronger response to priming, and this is most visible in their policy attitudes.

As an additional robustness check, Appendix Figure D.2 and Appendix Table D.1 estimate a flexible spline specification that allows treatment effects to vary nonlinearly with the Religiosity Index among religious respondents, while keeping the non-religious group separate. The results here are qualitatively consistent with the main analysis (Table 3) with more negative effects at lower religiosity and weaker or positive effects at higher religiosity; however, omnibus nonlinear tests are generally not statistically significant.

Table 3: Heterogeneous effects of religious priming by religiosity

	Allocations to profiles			Support for policies	
	All same-sex (1)	Male same-sex (2)	Female same-sex (3)	Same-sex marriage (4)	Child adoption (5)
Panel A: Subgroup treatment effects					
<i>Not religious</i>					
Religious-salient	-3.663 (8.246)	-1.855 (9.665)	-5.472 (8.757)	0.087 (0.115)	0.060 (0.114)
<i>N</i>	493	493	493	493	493
Control mean	215.653	210.718	220.587	2.487	2.351
<i>Less religious</i>					
Religious-salient	-8.264 (6.158)	-3.836 (7.337)	-12.692* (6.568)	-0.107 (0.083)	-0.193** (0.083)
<i>N</i>	755	755	755	755	755
Control mean	215.534	204.786	226.283	2.124	2.116
<i>More religious</i>					
Religious-salient	0.399 (6.214)	0.270 (7.395)	0.527 (6.695)	0.128 (0.082)	0.158* (0.083)
<i>N</i>	769	769	769	769	769
Control mean	207.313	200.582	214.044	1.915	1.852
Panel B: Interaction coefficients (reference = Not religious)					
Religious-salient $\times$ Less religious	-4.601	-1.982	-7.221	-0.194	-0.253*
Religious-salient $\times$ More religious	4.062	2.125	5.998	0.041	0.098
Control mean	215.653	210.718	220.587	2.487	2.351
Panel C: Formal heterogeneity tests					
<i>p</i> -value: Less vs More	0.323	0.693	0.160	0.044	0.003
<i>p</i> -value: Joint test of interactions	0.614	0.925	0.371	0.111	0.010
<i>N</i> (common pooled-model sample)	2017	2017	2017	2017	2017

Notes: Panel A reports subgroup treatment effects recovered from a pooled interaction model as linear combinations: Not religious = β_T , Less religious = $\beta_T + \beta_{T \times L}$, More religious = $\beta_T + \beta_{T \times M}$. Robust HC3 standard errors are in parentheses. Controls are the same as in the baseline specification. Panel B reports the interaction coefficients relative to the omitted Not-religious group. Panel C reports direct equality tests of subgroup treatment effects and a joint Wald test that both interaction terms are zero. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The heterogeneity analysis is exploratory, and conclusions depend on how the multiple-testing family is defined. In the robustness adjustment reported in Appendix Table D.2, only the negative effect on support for child adoption among the Less Religious remains significant at the 5 percent level. I therefore treat the heterogeneity evidence as suggestive overall, with the strongest support in policy attitudes — particularly child adoption⁷.

Overall, the religious prime does not have stable average effects across the main outcomes. Heterogeneity analysis shows that some effects of the religious prime emerge among the less religious: this group shows negative estimates — i.e., lower allocations to same-sex profiles and less support for pro-LGB policies — while the non-religious group shows little response and the more religious group does not show consistently negative effects. The

⁷An alternative interpretation of the heterogeneity pattern is that the prime reshuffles respondents along the religiosity scale rather than genuinely shifting behavior. Respondents in the treatment group who initially reported low religiosity may update upward upon reflection, while those who reported high religiosity may update downward, which would compress the distribution toward the middle and mechanically flatten the religiosity gradient in the treatment group. However, the randomization check on religion-related variables for the treatment and control groups (Appendix Table B.4) shows that their distributions are similar, providing limited evidence of systematic compression. Nevertheless, since these variables are measured as the treatment for the treated group, this alternative cannot be fully ruled out and should be kept in mind when interpreting the heterogeneity results.

pattern is also outcome-specific, with clearer differences for policy attitudes (especially child adoption) than for allocation outcomes. These differences are not statistically significant for the allocation outcomes. Finally, the strength of the heterogeneity results depends on the multiple hypothesis correction.

4 Conclusion

This paper examines the causal effect of the salience of religious identity on discrimination against sexual minorities. I conduct an online experiment in Russia, in which I use a priming technique to activate participants' religious identities and measure discriminatory behavior through incentivized allocation tasks and questions on LGB-related policy support. The results show that there is substantial discrimination against same-sex profiles. On average, participants allocate 77 RUB less to same-sex profiles than to different-sex profiles. Some characteristics - gender, age, and political ideology - are the strongest predictors of this discriminatory behavior.

I find no statistically significant average treatment effects of the religious prime on allocation decisions or policy preferences, even within the religious subsample. I then examine whether the prime operates differently across religiosity groups. The heterogeneity analysis suggests that the treatment effect is negative among religious respondents with moderate levels of religiosity, and near zero or positive among the most religious respondents, with the strongest evidence for attitudes towards policy, particularly child adoption. These patterns are consistent with the idea that the prime has the strongest effect among individuals for whom religion is meaningful but not a permanently salient part of their daily lives. However, the heterogeneity results are exploratory and sensitive to the choice of multiple hypothesis correction.

The interpretation and generalizability of these findings are subject to two limitations. First, the experiment is conducted in a specific cultural and institutional context — Russia, where Orthodox Christianity is dominant and anti-gay attitudes are widespread. Second, the religious battery used as the prime may not have been strong enough to induce a meaningful shift in religious identity salience.

Overall, this paper adds causal evidence to a literature that has largely relied on correlational approaches, and points to religious identity salience as a relevant channel through which religious affiliation may affect discriminatory behavior — one that calls for further investigation in other religious contexts, with larger samples and stronger priming interventions.

Appendix A Experimental Task Example

Figure A.1: Example of an allocation task between same-sex and different-sex profiles

HOW DO YOU WANT TO SPLIT ₺500 BETWEEN THESE TWO INDIVIDUALS?
Please attentively read the information about each of the individuals. Remember that both individuals have similar income.

How do I want to split the money?	
	
Name: <i>Ivan</i>	Name: <i>Maksim</i>
Age: <i>24</i>	Age: <i>23</i>
Partner's name: <i>Alina</i>	Partner's name: <i>Aleksandr</i>
Education: <i>Moscow State University, Economics</i>	Education: <i>Moscow P. I. Tchaikovsky Conservatory</i>
Job title: <i>Junior Researcher</i>	Job title: <i>Orchestra Artist</i>
₺250	₺250

Notes: This is a screenshot from the experiment, translated from Russian. The original tasks do not have the default amounts selected as in the screenshot.

Appendix B Sample Characteristics

Table B.1: Summary statistics

	Obs.	%	Census / benchmark %
<i>Sex</i>			
Male	1038	48	47
Female	1107	52	53
<i>Age groups</i>			
18–24	240	11	11
25–34	504	23	25
35–44	542	25	23
45–54	422	20	19
55–64	437	20	22
<i>Region</i>			
Central	673	31	28
Northwestern	276	13	6
Southern	213	10	11
North Caucasian	27	1	7
Volga	353	16	20
Ural	254	12	8
Far Eastern	72	3	9
Siberian	277	13	11
<i>Education</i>			
Incomplete secondary or below	12	1	21
Secondary general	180	8	17
Primary professional / secondary specialized	594	28	35
Student	115	5	3
Bachelors	716	33	6
Masters	488	23	17
Postgraduate	40	2	1
<i>Marriage status</i>			
Married	1180	55	45
Single	458	21	19
Widow	47	2	11
Divorced	208	10	10
Civil marriage	230	11	6
Hard to answer	22	1	–
<i>Sexual orientation</i>			
Heterosexual	1688	79	91
Homosexual	34	2	2
Bisexual	167	8	2
Other	92	4	–
Hard to answer	164	8	5
<i>Income groups</i>			
<20,000 RUB	139	6	–
20,000–49,900 RUB	626	29	–
50,000–79,900 RUB	598	28	–
80,000–99,900 RUB	343	16	–
100,000–199,900 RUB	352	16	–
>200,000 RUB	87	4	–
<i>Having children</i>			
No	659	31	–
Yes	1486	69	–
<i>Total</i>	2145	100	

Notes: Census data is retrieved by the author from official statistics web-page of the Federal State Statistics Service, <https://rosstat.gov.ru/folder/12781>.

Table B.2: Religious profile of the sample

	Obs.	%	Census / benchmark %
<i>Religion affiliation</i>			
Orthodoxy	1383	65.73	66
Catholicism	11	0.52	0
Protestantism	14	0.67	1
Islam	52	2.47	6
Buddhism	18	0.86	1
Judaism	5	0.24	0
No religion	500	23.76	14
Other	61	2.90	6
Hard to answer	60	2.85	6
<i>Subtotal</i>	2104	100	
<i>Importance of religion</i>			
Very important	159	8	
Fairly important	550	26	
Not very important	655	31	
Not at all important	528	25	
Hard to answer	211	10	
<i>Subtotal</i>	2103	100	
<i>Church attendance</i>			
A few times a week	20	1	
Once a week	54	3	
Two-three times a month	61	3	
Once a month	62	3	
A few times a year	349	17	
Once a year	225	11	
Less than once a year	494	23	
Never	755	36	
Hard to answer	84	4	
<i>Subtotal</i>	2104	100	
<i>Praying frequency</i>			
Often	314	15	
Sometimes	381	18	
Hardly ever	362	17	
Only in times of crisis	303	14	
Never	626	30	
Hard to answer	117	6	
<i>Subtotal</i>	2103	100	

Notes: Different subtotals reflect item-level missing values. Overall sample size is $N = 2145$. Religion affiliation and church attendance are observed for $N = 2104$; importance of religion and praying frequency are observed for $N = 2103$. Census shares are based on (non-official) Russian Public Opinion Research Center (VCIOM) (2021).

Table B.3: Randomization check: Non-religion variables

Variable	Control	Religious-salient	Diff (T-C)	S.E.	p-value
<i>Age and Gender</i>					
Age	41.674	42.251	0.576	0.540	0.286
Female	0.513	0.519	0.006	0.022	0.780
Male	0.487	0.481	-0.006	0.022	0.780
<i>Region</i>					
Central	0.307	0.321	0.014	0.020	0.495
Northwestern	0.129	0.129	-0.000	0.014	0.993
Southern	0.099	0.100	0.001	0.013	0.948
North Caucasian	0.015	0.010	-0.005	0.005	0.332
Volga	0.171	0.158	-0.012	0.016	0.444
Ural	0.112	0.125	0.013	0.014	0.354
Far Eastern	0.034	0.034	-0.000	0.008	0.997
Siberian	0.134	0.124	-0.010	0.014	0.474
<i>Employment</i>					
Unemployed	0.188	0.200	0.012	0.017	0.485
Employed	0.757	0.760	0.004	0.018	0.831
Student	0.055	0.039	-0.016*	0.009	0.082
<i>Income</i>					
< 20,000 RUB	0.065	0.064	-0.001	0.011	0.926
20,000 - 49,900 RUB	0.289	0.295	0.005	0.020	0.786
50,000 - 79,900 RUB	0.293	0.265	-0.028	0.019	0.145
80,000 - 99,900 RUB	0.147	0.172	0.025	0.016	0.114
100,000 - 199,900 RUB	0.167	0.161	-0.006	0.016	0.720
> 200,000 RUB	0.038	0.043	0.005	0.009	0.588
<i>Children</i>					
No children	0.318	0.296	-0.022	0.020	0.276
Have children	0.682	0.704	0.022	0.020	0.276
<i>Political Identity</i>					
Liberal	0.198	0.195	-0.003	0.017	0.862
Conservative	0.305	0.325	0.020	0.020	0.314
Apolitical(pooled)	0.459	0.454	-0.005	0.022	0.813
<i>Education</i>					
Low education	0.097	0.082	-0.015	0.012	0.224
High education	0.903	0.918	0.015	0.012	0.224
<i>Sexual Orientation</i>					
Heterosexual	0.789	0.785	-0.004	0.018	0.801
Homosexual	0.017	0.015	-0.002	0.005	0.728
Bisexual	0.071	0.085	0.014	0.012	0.229
Other	0.045	0.041	-0.004	0.009	0.667
Hard to answer	0.078	0.075	-0.004	0.011	0.741
<i>Social Contact</i>					
Having a friend with homosexual sexual orientation	0.247	0.283	0.036*	0.019	0.067
<i>N</i>	1,072	1,073			

Notes: Diff (T-C) is treatment minus control. p-values come from OLS checks with HC3 robust standard errors. Low education includes incomplete secondary education or secondary general education; high education includes primary professional/secondary specialized education, incomplete higher education/student status, Bachelor's, Master's, or postgraduate education. Omnibus joint-balance test: $F(30, 2043) = 0.695$, p-value = 0.891. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Randomization check: religion variables

Variable	Control	Religious-salient	Diff (T-C)	S.E.	p-value
<i>Religion affiliation (overall chi-square $p = 0.6817$)</i>					
Orthodoxy	0.6541	0.6604	0.0064	0.0207	0.758
Catholicism	0.0029	0.0075	0.0046	0.0031	0.144
Protestantism	0.0058	0.0075	0.0016	0.0035	0.641
Islam	0.0213	0.0280	0.0067	0.0068	0.324
Buddhism	0.0078	0.0093	0.0016	0.0040	0.694
Judaism	0.0019	0.0028	0.0009	0.0021	0.685
No religion	0.2510	0.2248	-0.0262	0.0186	0.159
Other	0.0262	0.0317	0.0056	0.0073	0.447
Hard to answer	0.0291	0.0280	-0.0011	0.0073	0.881
<i>Religion importance (overall chi-square $p = 0.0949$)</i>					
Very important	0.0660	0.0849	0.0189*	0.0115	0.100
Fairly important	0.2580	0.2649	0.0069	0.0192	0.718
Not very important	0.2987	0.3237	0.0250	0.0202	0.217
Not at all important	0.2658	0.2369	-0.0288	0.0189	0.128
Hard to answer	0.1115	0.0896	-0.0220*	0.0131	0.094
<i>Church attendance (overall chi-square $p = 0.6068$)</i>					
A few times a week	0.0078	0.0112	0.0034	0.0042	0.415
Once a week	0.0262	0.0252	-0.0010	0.0069	0.888
Two-three times a month	0.0262	0.0317	0.0056	0.0073	0.447
Once a month	0.0262	0.0326	0.0065	0.0074	0.379
A few times a year	0.1609	0.1707	0.0099	0.0162	0.544
Once a year	0.1017	0.1119	0.0102	0.0135	0.449
Less than once a year	0.2277	0.2416	0.0139	0.0185	0.453
Never	0.3818	0.3368	-0.0450**	0.0209	0.031
Hard to answer	0.0417	0.0382	-0.0034	0.0086	0.689
<i>Prayer frequency (overall chi-square $p = 0.2410$)</i>					
Often	0.1455	0.1530	0.0075	0.0156	0.630
Sometimes	0.1639	0.1978	0.0338**	0.0168	0.044
Hardly ever	0.1775	0.1670	-0.0105	0.0165	0.523
Only in times of crisis	0.1445	0.1437	-0.0009	0.0153	0.955
Never	0.3046	0.2910	-0.0135	0.0200	0.499
Hard to answer	0.0640	0.0476	-0.0164	0.0100	0.101

Notes: This table reports the balance between the religion-related variables. Diff (T-C) is estimated from level-specific linear probability models with HC3 robust standard errors. Overall p-values in panel headers are chi-square tests of distributional equality by treatment status. Joint balance test across all raw religion variables: $F(25, 2077) = 0.815$, $p = 0.726$, $N = 2103$. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

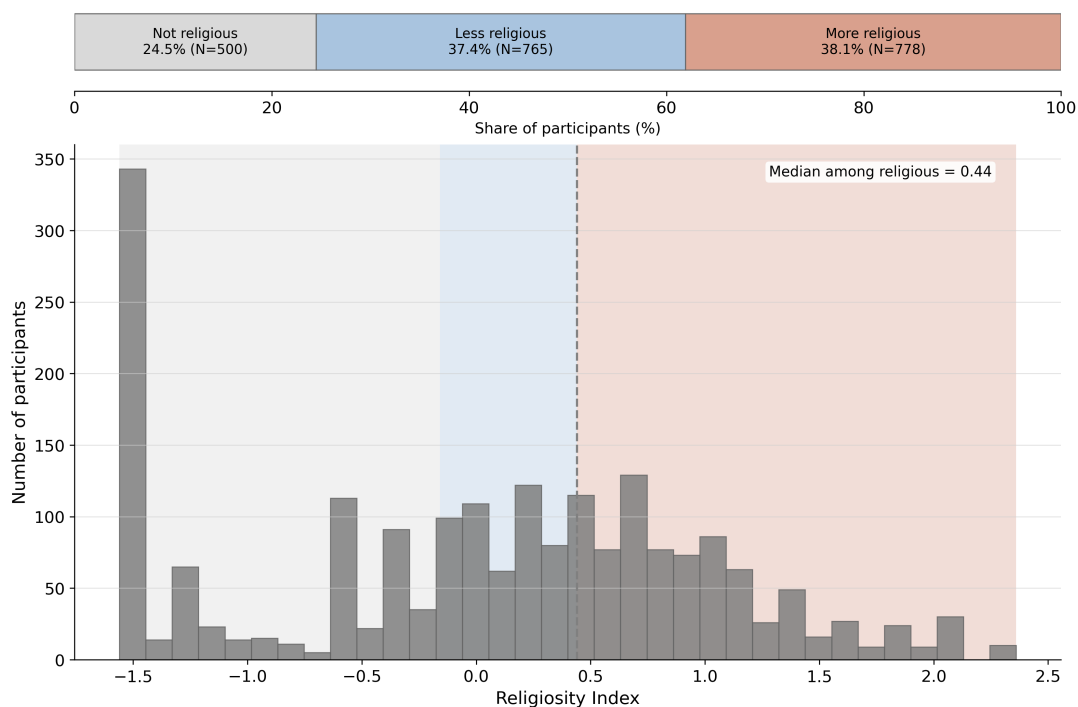
Appendix C Religiosity Index

Table C.1: Correlations among the components of the Religiosity Index

	Religious affiliation	Church attendance	Prayer frequency	Religious importance
Religious affiliation	1.0000			
Church attendance	0.4326***	1.0000		
Prayer frequency	0.5212***	0.5906***	1.0000	
Religious importance	0.5176***	0.5825***	0.6832***	1.0000

Notes: The table reports pairwise Pearson correlations among the four components used to construct the Religiosity Index: religious affiliation, church attendance, prayer frequency, and the subjective importance of religion. Religious affiliation is coded as an indicator for being religiously affiliated, with non-substantive responses pooled to the lowest category in the main specification. Church attendance, prayer frequency, and religious importance are coded so that higher values indicate higher religiosity. Correlations are computed using pairwise-complete observations. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

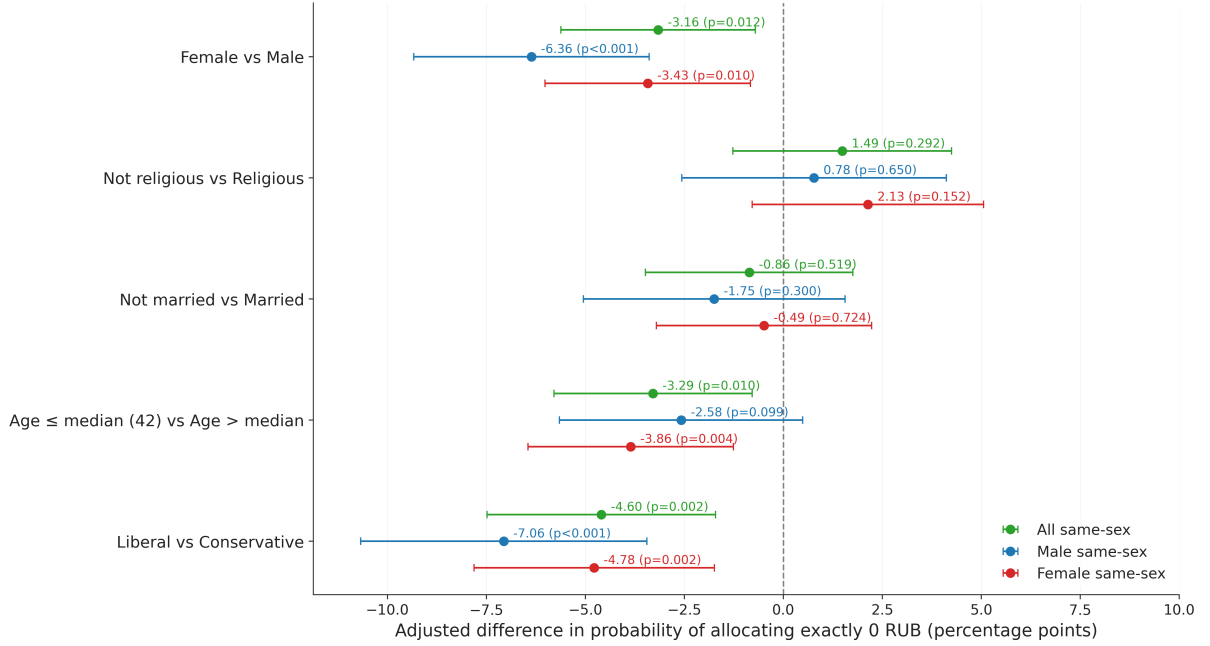
Figure C.1: Distribution of the Religiosity Index



Notes: The figure shows the distribution of the Religiosity Index. Higher values indicate higher religiosity. “Not religious” includes respondents whose raw religious affiliation is “No religion.” Among affiliated respondents, the dashed line marks the median religiosity index used to distinguish the less religious and more religious groups.

Appendix D Additional Results

Figure D.1: Subgroup differences in zero allocations to same-sex profiles



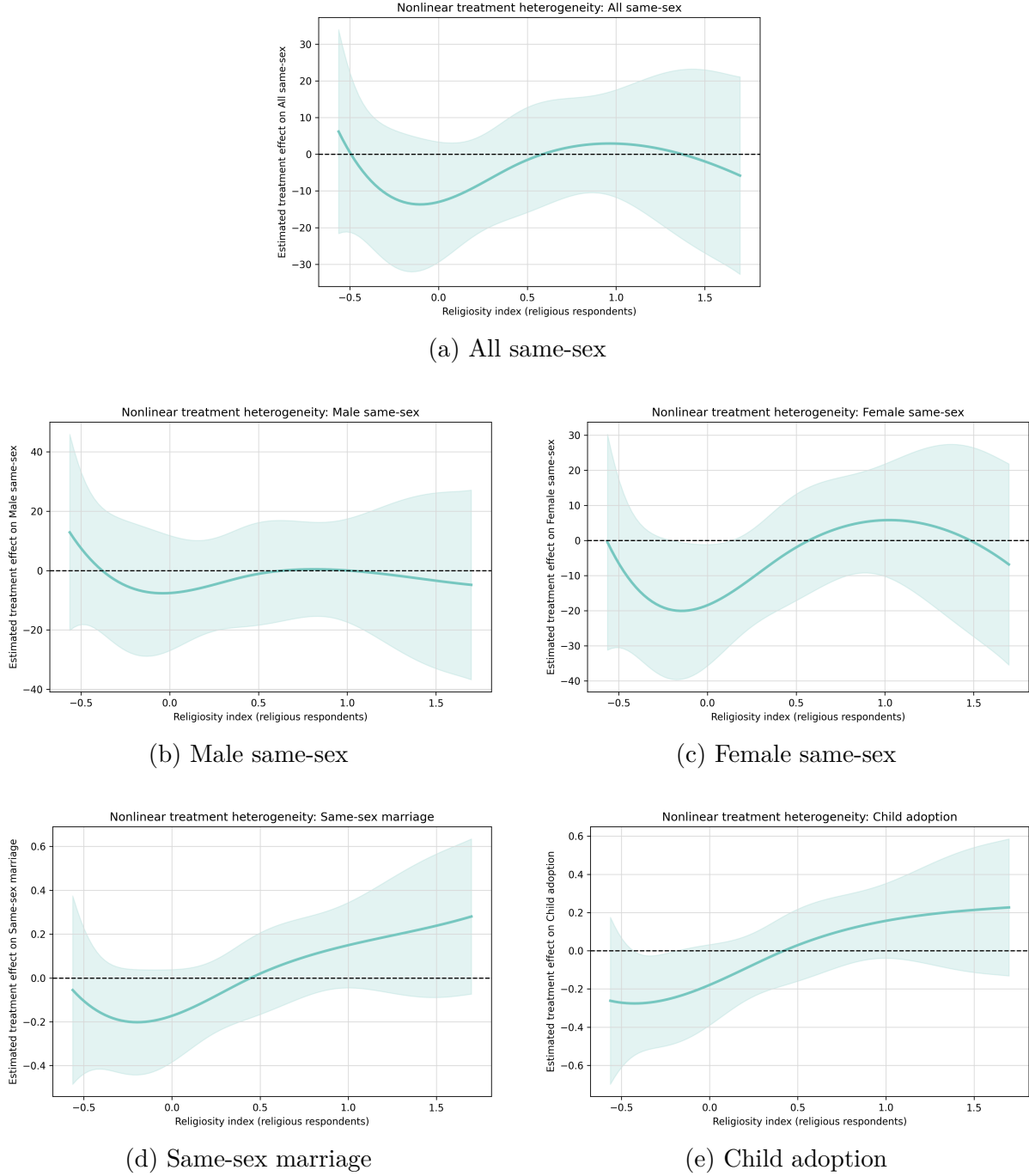
Notes: Dots report adjusted subgroup differences in the probability of allocating exactly 0 RUB to same-sex profiles, measured in percentage points. Whiskers denote 95% confidence intervals based on HC3 robust standard errors. The dashed vertical line marks zero effect. Text labels report coefficient estimates and p -values. Each row is estimated from a separate OLS specification with controls aligned to the main heterogeneity model: gender, age, education, income, marital status, number of children, political identity, prior belief about the biological causes of sexual orientation, sexual orientation, and having a homosexual friend; the subgroup-defining covariate is omitted in each row to avoid collinearity. Reference groups are Male, Religious, Married, Age > median, and Conservative.

Table D.1: Spline-based tests of nonlinear treatment heterogeneity

Outcome	N	Not religious	Overall p	BH-FDR q	Affiliated spline p	BH-FDR q
All same-sex	2017	-3.619 (8.248)	0.805	—	0.680	—
Male same-sex	2017	-1.865 (9.665)	0.962	0.962	0.910	0.910
Female same-sex	2017	-5.373 (8.759)	0.537	0.962	0.394	0.788
Same-sex marriage	2017	0.089 (0.115)	0.190	0.190	0.133	0.133
Child adoption	2017	0.062 (0.114)	0.134	0.190	0.090	0.133

Notes: This table reports omnibus tests from the spline-based heterogeneity specification shown graphically in Figure D.2. The model keeps the non-religious group separate and allows the treatment effect to vary flexibly with the Religiosity index among religious respondents using a cubic spline. “Not religious effect” reports the estimated treatment effect for respondents with no religious affiliation, with HC3 robust standard errors in parentheses. “Overall p ” is a joint test that the treatment effect does not vary between the non-religious group and religious respondents and does not vary with religiosity among the religious. “Affiliated spline p ” is a joint test that all treatment-by-spline interaction terms equal zero among religiously affiliated respondents. Benjamini-Hochberg adjustments are applied within outcome blocks, following the paper’s preferred family definition: male and female same-sex allocations form the allocation block, while same-sex marriage and child adoption form the policy block. The pooled all same-sex allocation outcome is treated as a descriptive summary and is therefore not included in the multiple-testing adjustment. The estimated curves are qualitatively consistent with Table 3, but the omnibus tests do not provide statistically significant evidence of nonlinear heterogeneity after adjustment.

Figure D.2: Spline-based treatment heterogeneity by religiosity



Notes: The figure reports treatment effects from a flexible spline specification that keeps the non-religious group separate and allows the treatment effect to vary nonlinearly with the Religiosity index among religious respondents. The solid line shows the estimated treatment effect of the Religious-salient prime; shaded bands are 95% confidence intervals based on HC3 robust standard errors. The horizontal dashed line marks zero effect. Because the religiosity index is defined only for religious respondents, the horizontal axis is shown only for that subsample; the treatment effect for the non-religious group is estimated separately and reported in Appendix Table D.1. Across outcomes, the estimated curves are qualitatively consistent with the grouped heterogeneity analysis in Table 3: effects are more negative at lower values of religiosity and become weaker or positive at higher values. However, the omnibus tests reported in Appendix Table D.1 are not statistically significant after multiple-hypothesis adjustment.

Table D.2: Multiple-hypothesis correction for heterogeneous treatment effects by religiosity

Outcome	Coef.	S.E.	Raw p	BH-FDR q
<i>Panel A: Not Religious</i>				
All same-sex	-3.663	(8.246)	0.657	—
Male same-sex	-1.855	(9.665)	0.848	0.848
Female same-sex	-5.472	(8.757)	0.532	0.848
Same-sex marriage	0.087	(0.115)	0.449	0.599
Child adoption	0.060	(0.114)	0.599	0.599
<i>Panel B: Less Religious</i>				
All same-sex	-8.264	(6.158)	0.180	—
Male same-sex	-3.836	(7.337)	0.601	0.601
Female same-sex	-12.692	(6.568)	0.053	0.107
Same-sex marriage	-0.107	(0.083)	0.198	0.198
Child adoption	-0.193	(0.083)	0.021	0.041
<i>Panel C: More Religious</i>				
All same-sex	0.399	(6.214)	0.949	—
Male same-sex	0.270	(7.395)	0.971	0.971
Female same-sex	0.527	(6.695)	0.937	0.971
Same-sex marriage	0.128	(0.082)	0.118	0.118
Child adoption	0.158	(0.083)	0.058	0.116

Notes: This table reports raw and multiple-hypothesis-adjusted p -values for the subgroup treatment effects in Table 3. Benjamini–Hochberg false-discovery-rate (q) values are computed separately within each religiosity subgroup and within construct block. For the allocation block, the correction is applied to the two sex-specific allocation outcomes only (Male same-sex and Female same-sex). The pooled "All same-sex" outcome is reported for completeness but is not included in the correction because it is mechanically derived from the male and female same-sex allocation outcomes. For the policy block, the correction is applied to the two policy outcomes (Same-sex marriage and Child adoption). Thus, the adjusted q -values shown here correspond to a narrower, construct-based family definition and do not correspond to a single correction across all outcomes in Table 3. Under this family definition, the negative effect on support for child adoption in the Less Religious subgroup remains significant at the 5 percent level after adjustment ($q = 0.041$).

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